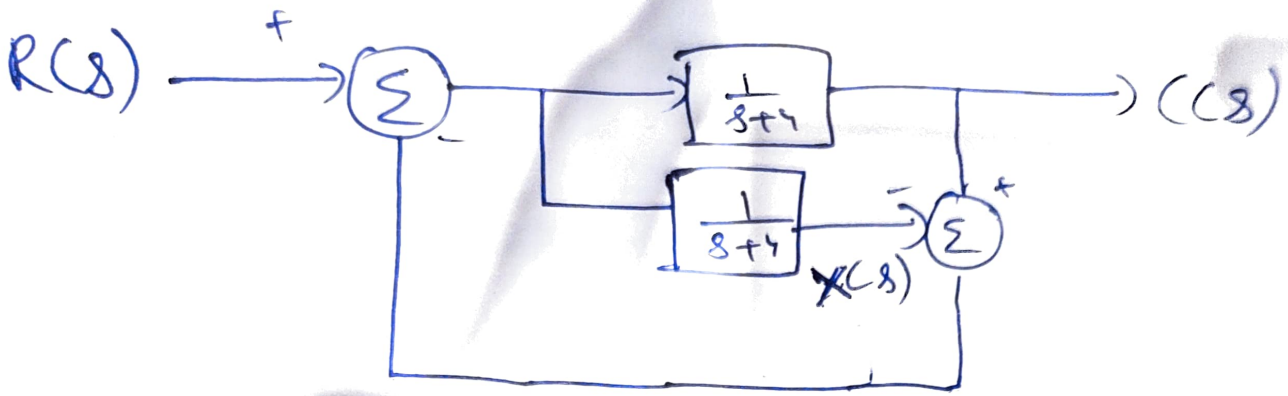


Let E(s) be the output of first summation and B(s) be the feedback signal from second summation. At first summing junction

$$E(s) = R(s) - B(s) \quad - (1)$$

$$C(s) = \frac{E(s)}{s+4} \quad - (2)$$

Now, $X(s)$ also $= \frac{E(s)}{s+4}$

$$B(s) = C(s) - X(s) = 0 \quad -(3)$$

Substituting (2) and (3) in (1),

$$C(s) = \frac{R(s)}{s+4}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{1}{s+4}$$